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5th Sem / Arch
Subject :- Reinforced Cement Concrete (RCC)

M.M. : 100

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Write the assumptions in Limit State of collapse in flexure(bending).
- Q.37 Find the M.O.R. of a R.C.C beam 350 mm x 550 mm (Effective), reinforced with 3 bars of 20 mm diameter. The permissible stresses in steel and concrete are 230N/mm^2 and 7N/mm^2 respectively. Take modular ratio as 13.33.
- Q.38 Write in detail about different systems of pre-stressing with sketches.

- Q.6 Resisting moment constant (R) depends upon-
- a) Grade of concrete and grade of steel
 - b) Beam dimensions
 - c) percentage of tensile steel
 - d) all of the above

Q.7 F.O.S of steel is-

- a) 2 b) 3
- c) 1.5 d) 1.78

Q.8 The spacing of stirrups near the supports is

- a) minimum b) maximum
- c) 400 mm d) 350 mm

Q.9 HYSD stands for

- a) High yielding strength deformed steel
- b) High yielding stress demand steel
- c) High yellow strength demand steel
- d) High yield strong deformed strength

Q.10 In a slab floor system the end beams are called

- a) T-beam b) P-beam
- c) L-beam d) none of these

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 WSM stands for_____.

Q.12 The portion of the T beam below the slab is called_____.

Q.13 Bars are preferred over mild steel bars.

Q.14 The axis where stresses are zero in a R.C.C beam is called_____.

Q.15 Partial safety factors for concrete is_____.

Q.16 Failure of an over reinforced beam section is sudden. (T/F)

Q.17 Mild steel is used in construction of steel column. (T/F)

Q.18 The neutral axis always passes through flange of a T-beam. (T/F)

Q.19 In WSM no factor of safety is used for loads. (T/F)

Q.20 LSM is more economical.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What is Mild steel and what are uses of mild steel in construction.

Q.22 Explain grade of concrete. What is M-25?

Q.23 What do you mean by TMT bars and what are its uses?

Q.24 Write the drawbacks of WSM.

Q.25 What is neutral axis? Draw the sketch showing neutral axis in balanced and unbalanced beam sections.

Q.26 Why is working stress method called deterministic method and limit stress method called non deterministic method?